Understanding Retail Customers Through RFM Clustering

A Python study of purchase activity, customer value and targeted marketing

Submitted by	Academic details
Kartik Kansal Registration No.: RA2411028030012	B.Tech - Computer Science and Engineering SRMIST Delhi NCR

1. Aim and scope

The aim is to divide retail customers into four groups using their transaction history. Instead of treating all buyers alike, the project measures how recently they purchased, how often they placed orders and how much they spent. The resulting groups provide a practical starting point for customer retention and marketing decisions.

2. Data used in the study

The supplied OnlineRetail.csv contains 541,909 transaction rows and eight columns: InvoiceNo, StockCode, Description, Quantity, InvoiceDate, UnitPrice, CustomerID and Country. After applying the validity rules, 397,884 rows remain, representing 4,338 identified customers.

The analysis uses the complete supplied file and does not filter by country. The final valid transaction occurred on 9 December 2011 at 12:50. A reference date of 10 December 2011 at midnight is used to calculate recency.

3. Preparing reliable purchase records

InvoiceDate is converted to a datetime value. Records are excluded if the customer identifier or date is missing, the invoice number starts with C, or quantity or unit price is zero or negative. Customer identifiers are converted to integers, and a Sales column is calculated as Quantity multiplied by UnitPrice.

Duplicate rows are retained, matching the original analysis. Missing descriptions do not cause a record to be removed because descriptions are not required for the RFM calculations.

From Transactions to Customer Groups

4. Constructing the customer profile

Feature	Calculation	Meaning
Recency	Whole days between the reference date and the latest purchase	Smaller values indicate more recent activity.
Frequency	Number of distinct invoice numbers per customer	Measures orders, rather than individual product lines.
Monetary	Sum of Sales for each customer	Measures observed sales value, rather than profit.

Each customer appears once in the resulting RFM table. Recency uses elapsed whole days, so a purchase on the final day can have a recency of zero when fewer than 24 hours separate it from the reference date.

5. Model design and implementation

The three RFM features are transformed using log(1 + x) to reduce the influence of strongly skewed values. StandardScaler then standardises the transformed features. K-means is fitted on this scaled matrix with four clusters, random_state = 42 and n_init = 10. Cluster numbers identify groups; they do not represent a ranking.

The rewritten Python implementation separates loading, cleaning, RFM calculation, clustering, summarisation and plotting into functions. Its default input is the user-provided Windows CSV path. It exports customer_segments.csv and segment_summary.csv and saves a two-panel chart as customer_cluster_charts.png.

6. Observed segment characteristics

Cluster	Customers	Median recency (days)	Median orders	Median spend (GBP)
0	1,622	173	1	296.81
1	810	16	2	470.12
2	713	8	10	3,729.34
3	1,193	54	4	1,341.65

Cluster 2 has the strongest combination of recent activity, frequent ordering and high typical spend. Cluster 0 contains the largest customer count but has the longest median time since the last order. Clusters 1 and 3 occupy different middle positions: recent occasional purchases and established repeat purchasing, respectively.

7. Validation of the rewritten code

Both implementations were executed on the supplied CSV in the same environment. Full customer-level RFM values, cluster assignments and segment summary tables matched. The silhouette score was 0.337. This indicates overlapping groups rather than sharply isolated customer types; the score alone does not establish an optimal number of clusters.

Validation environment: pandas 2.2.3, NumPy 2.3.5 and scikit-learn 1.8.0. Matching package versions supports reproducibility; cluster labels can differ across environments.

Turning the Results into Actions

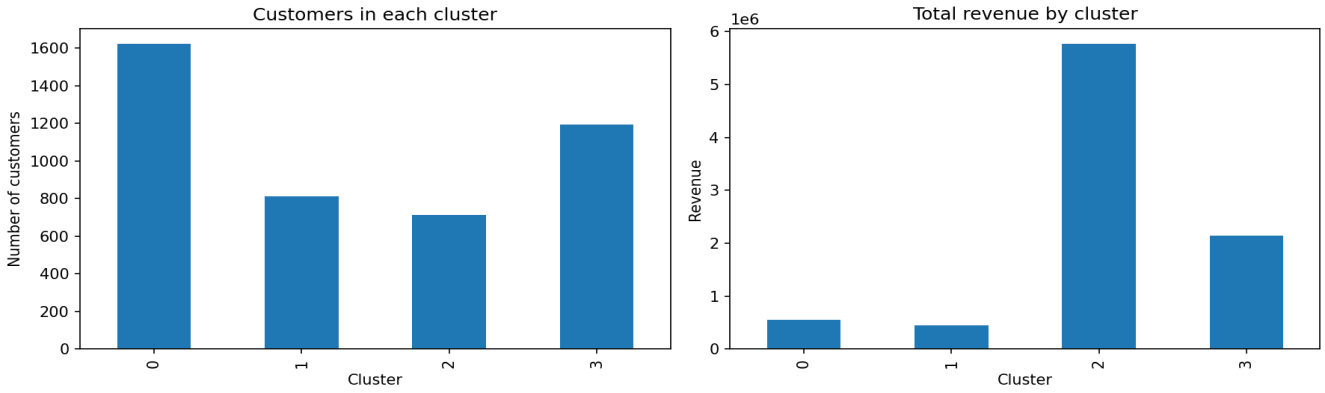


Figure 1. Customer counts and total observed sales by cluster, generated by the rewritten program.

8. Revenue concentration and marketing priorities

Cluster 2 accounts for 16.4% of customers and 64.7% of sales. Cluster 3 contributes another 24.1% of sales from 27.5% of customers. Together, these two groups generate 88.8% of observed sales. These figures suggest that retaining active, valuable buyers deserves particular attention.

Group interpretation	Recommended response
Cluster 0 | Dormant / at risk 37.4% of customers; 6.2% of sales	Test a low-cost reactivation message with relevant products. Monitor reactivation and stop repeated contact when customers remain unresponsive.
Cluster 1 | Recent / occasional 18.7% of customers; 5.0% of sales	Use post-purchase guidance and complementary recommendations to encourage a second order.
Cluster 2 | Champions / VIP 16.4% of customers; 64.7% of sales	Prioritise retention through service, loyalty benefits and early access. Measure repeat purchases and retained sales.
Cluster 3 | Regular / growth potential 27.5% of customers; 24.1% of sales	Offer relevant bundles and replenishment reminders. Track order frequency and average basket value.

9. Limitations and possible extensions

The dataset ends in 2011, so the groups describe historical purchasing behaviour. Excluding cancelled and negative transactions means monetary value reflects qualifying positive sales, not net revenue after returns. Customers without identifiers are excluded. Four clusters were selected in advance and were not proven optimal in this study.

Future work could compare alternative cluster counts, evaluate stability over time and test campaigns against a holdout group. Marketing recommendations in this report are proposed actions; their business impact has not been experimentally measured.

10. Conclusion

RFM clustering converts a large transaction file into interpretable customer profiles. The rewritten implementation preserves the original numerical results while organising the workflow into reusable functions. Its main finding is a strong concentration of sales in the active, higher-value customer groups, supporting differentiated retention, growth and reactivation strategies.

Materials used: supplied OnlineRetail.csv; Module_4_Customer_Segmentation.ipynb; verified outputs of customer_segmentation_rewritten.py. All tables and findings are derived from these supplied project materials.